

where q is the velocity pressure ($\frac{1}{2}\rho V^2$), ρ is the air density, V the wind speed, μ the viscosity of air, and L a linear dimension fixing the scale, taken as the diameter in the case of cylinders. The expression applies only to geometrically similar bodies, for example, to cylinders of infinite length or of constant ratio of length to diameter. The pressure, p , may be regarded as consisting of two parts, the static pressure,² p_s , which would exist in the absence of the cylinder, and the pressure change, $p - p_s$, caused by the presence of the cylinder. In a natural wind p_s is the atmospheric pressure. $p - p_s$ is the quantity usually measured, corresponding to the common practice of using "gauge" pressure in dealing with pressures in steam boilers, compressed air tanks, etc., rather than absolute pressure. The pressure difference could be measured in any convenient units, but there are advantages in using the velocity pressure, $q = \frac{1}{2}\rho V^2$, as the unit.

In the first place $\frac{p - p_s}{q}$ is in many cases³ independent of the wind speed and the size of model, so that from a single value of it for any given shape of body the pressure at the corresponding point on a similar body of any size at any wind speed can be readily computed with the aid of a table of velocity pressures. In many other cases the variation of $\frac{p - p_s}{q}$ with the Reynolds number $\frac{VL\rho}{\mu}$ is small. The ratio is a pure number independent of the units used so long as the pressures are all measured in the same units.

TABLE 2.—Velocity pressures at several wind speeds

True wind speed (miles per hour)	Velocity pressure (lbs./ft. ²)	True wind speed (miles per hour)	Velocity pressure (lbs./ft. ²)
5	0.064	65	10.81
10	.256	70	12.53
15	.575	75	14.39
20	1.023	80	16.37
25	1.600	85	18.48
30	2.302	90	20.72
35	3.133	95	23.08
40	4.092	100	25.58
45	5.179	105	28.20
50	6.394	110	30.95
55	7.737	115	33.83
60	9.208	120	36.83

NOTE.—Velocity pressure in [lbs./ft.² = 0.001189 ($V \times \frac{22}{15}$)² where V is the true speed in miles per hour, and the density is that at 15° C.

The engineer is in most cases more directly concerned with the resulting forces than with the pressures themselves and their distribution. The force on any element, dA , is equal to $p dA$ and acts in the direction of the normal to the surface.⁴ If θ is the angle made by the surface element, s , with the wind direction, the component of force in that direction, F_s , is given by $\int \int p dA, \sin \theta$. Since p_s is

² The term static pressure is used to indicate the pressure which would be measured by a pressure gauge moving with the air and, therefore, "static" with respect to the air. In actual practice the measurement is made by means of holes in the side of a closed tube, the axis of which is parallel to the wind direction. The form of the tube is such that the air flows smoothly past the holes.

³ B. S. Sci. Paper No. 523, Bull. 20, p. 697.

⁴ Excluding frictional effects, which are of importance only in special cases.